

Issue #2: The website attracts traffic that never converts.

If a website has countless numbers of online visitors flocking on a daily basis but none of them ever convert – i.e., turn to targeted leads or even sales – then all these hits that the site is accumulating are just worthless statistics that wouldn't do the business any good. Even if one has a million website hits that result in zero conversions, it would end up with zero sales – an extremely terrible scenario for any business.

Suggested solution: Creation of targeted traffic

The first step is to generate targeted traffic. Targeted traffic is “qualified” or “quality” traffic. What this means is that site visitors coming through to the website or landing pages are most likely actively looking for the products, services, and other information that the company's website already has. This should be the goal of the website and the overall digital marketing efforts. The following reasons explain why:

Targeted Traffic Provides Better Chances for Conversion.

Targeted traffic offers better chances for conversion as the majority of online visitors are seeking information that the website may supply, and have a greater chance of purchasing the products or services should they cater to whatever needs they may have. Even if one only has a trickle of website traffic, if these visitors can be converted into either leads or sales, then that would be traffic worth receiving.

Targeted traffic builds the brand reputation.

People online are constantly seeking huge amounts of information for whatever purposes or needs they may have on a daily basis. More often than not online users get directed to a website, either through a natural search engine query or clicking on what they thought was a relevant PPC ad, only to find information or content that really does not provide the answers they need. Many simply get frustrated from all these and forever shun these websites that they have now marked as useless.

If one is generating the right kind of targeted traffic to the website – the kind that is actively searching for content about certain kinds of products or services – and if the website manages to deliver the content they need, one will end up with happy, satisfied visitors who'll have no qualms of going back to the website on a regular basis. This would truly be a big boost to the brand's online reputation, which will be catapulted to even greater heights once these satisfied web visitors start telling other people about the brand, product, or services.

Targeted traffic helps increase the page rankings.

Targeted traffic is quality traffic which not only increases the sales but improves the branding and online reputation as well. With high level of credibility among these targeted users and the site will eventually be branded as an authority and reliable source in the respective niche. This will give the website more opportunities for generating backlinks from relevant websites.

As discussed before regarding SEO, a good flow of targeted traffic reinforced by incoming links is a surefire ingredient in helping the website rank higher in search

engine results pages (SERPs). These are the kinds of factors search engines like Google love, and they will reward you with good rankings.

18.7 CASE STUDIES

18.7.1 LENOVO

Lenovo is one of the largest manufacturers of computer hardware, namely desktop computers, laptops, and related technology products. Lenovo's marketing team formulated a content marketing strategy to first create awareness and then build trust among their target audience, which mainly comprises IT professionals. The aim was to help them make informed buying decisions and to keep Lenovo at top-of-mind recall while making those decisions.

18.7.1.1 Campaign Objective

Create content to educate Lenovo's target audience and distribute it across multiple channels.

18.7.1.2 Method and Execution

To execute its content marketing strategy, Lenovo developed a digital content hub called "Tech Revolution" to deliver technical news and technology information to IT decision-makers across Asia Pacific region.

They were able to successfully engage with their target audience through this content marketing hub. The leads generated via Tech Revolution were routed to their website, based on which sales executives were able to readily connect with consumers.

18.7.1.3 Results

The content marketing strategy of Lenovo was hugely successful in establishing the company as a thought leader. The digital content hub had more than 250 articles with 34 million impressions, 308 link clicks, and was visited by 170,000 new web users. In terms of financials, \$300 million in sales was attributable to this initiative alone.

18.7.2 SALESFORCE

Customer relationship management (CRM) is among the most competitive domains in the IT industry. Salesforce enjoys a premier position, being the world's largest vendor of CRM applications.

While Salesforce was successful in most global locations, it had some issues maintaining steady traffic and leads in its UK business.

Salesforce decided to change their content marketing strategy to achieve business growth in search-and-paid traffic in UK.

18.7.2.1 Campaign Objective

To deploy content marketing to entertain and engage target audience in UK to achieve business growth in the region.

18.7.2.2 Method and Execution

Salesforce created the following digital assets:

- Video on sales and marketing which was highly entertaining and fun.
- The world's first stop-motion slide share.
- A Prezi presentation to showcase a customer success story.

18.7.2.3 Results

Salesforce's team in UK achieved some dramatic results as a result of this content marketing campaign. They enjoyed 80% increase in web traffic, over 10,000 downloads of their customized e-book, approximately 6,500 email newsletter signups, and a staggering 2,500% rise in traffic from social media sites.

18.8 CONCLUSION

For the issues pertaining to when the published content failed to generate new leads, it was found that cause was that the published content didn't help people solve the problems they face. It must always be kept in mind while developing content that the content that doesn't solve problems will not be searched for on the internet.

Hence, content developers need to focus on creating content that adds value or solves problems and not just writing to sell. In addition, to get more leads, organizations should develop efficient CTAs that compel users and potential customers to avail services and purchase products.

To battle the information saturation present in an industry, an organization must strive to offer its customers a fresher perspective regarding the topics of interest if they do not wish to lose any audience. This can be achieved by writing and developing targeted content, because it is the only form of content creation strategy which guarantees relevance to every reader (Khang et al., 2023b).

For the issues related to published content that did not create brand awareness, it was concluded that following the prescribed procedure can ensure a greater probability of generating the in trended brand awareness (Misra et al., 2023).

Furthermore, two primary reasons were identified that damaged the created awareness about the brand or worse, created negative awareness about the brand. This may have happened due to pieces of content that were created and published with the single aim of making a sale. Such content lacks elements which make it engaging (Morris et al., 2023).

The customers of today can easily distinguish between a company that cares and strives to solve problems and are most likely to lean toward a company that not only sells but also offers after-sales service. This persona can be achieved by simply conveying the customers with the best possible information and solutions (Vrushank, Khang & Rani, 2023).

When the content fails to establish the publishing organization as the expert of its domain, the process of developing content must be tweaked in the following manner: Create content that attracts huge amounts of traffic from all over the internet (Tailor, Pareek & Khang, 2022).

As the traffic increases, the website will begin advancing in the ranking at search engine results and thus showcasing itself as the expert of the domain. The higher up the website ranks, the more credible it is perceived to be. This can be achieved by generating effective keywords and creating, developing, publishing, and managing the content of quality and relevance (Khang et al., 2022b).

REFERENCES

- Babasaheb, J., Sphurti, B., & Khang, A., "Design of Competency Models in the Human Capital Management System," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023a), pp. 32–50. CRC Press. <https://doi.org/10.1201/9781003357070-3>
- Babasaheb, J., Sphurti, B., & Khang, A., "Industry Revolution 4.0: Workforce Competency Models and Designs," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023b), pp. 14–31. CRC Press. <https://doi.org/10.1201/9781003357070-2>
- Berger, J., & Milkman, K., What makes online content viral? *Journal of Marketing Research*, 2012, XLIX, 192–205. <https://marketing.wharton.upenn.edu/files/?whdmsaction=public:main.file&fileID=3461>
- Bhambri, P., Rani, S., Gupta, G., & Khang, A., *Cloud and Fog Computing Platforms for Internet of Things* (2022). CRC Press. <https://doi.org/10.1201/9781032101507>
- Cialdini Robert, Simon and Schuster (2016). Pre-Suasion: A Revolutionary Way to Influence and Persuade. Retrieved from <https://books.google.com.vn/books?hl=en&lr=&id=GI7tDAAQBAJ>
- Copyblogger (2013). Content Marketing: How to Build an Audience that builds Your Business. <http://www.copyblogger.com/content-marketing/>
- JODI HARRIS (2023). The One Thing You Need To Include in Your Newsletter, "5 Essential Elements for a Great Newsletter". Retrieved from <https://contentmarketinginstitute.com/articles/five-essential-elements-to-a-great-newsletter/>
- Gupta, S.K., Somani, P., Dixit, C.K., Pathak, A., & Khang, A., "Data Mining Processes and Decision-Making Models in Personnel Management System," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023), pp. 89–112. CRC Press. <https://doi.org/10.1201/9781003357070-6>
- Hajimahmud, V.A., Khang, A., Hahanov, V., Litvinova, E., Chumachenko, S., & Alyar, A.V., "Autonomous Robots for Smart City: Closer to Augmented Humanity," *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) (2022). CRC Press. <https://doi.org/10.1201/9781003252542-7>
- Jain, P., Tripathi, V., Malladi, R., & Khang, A., "Data-Driven AI Models in the Workforce Development Planning," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023), pp. 179–198. CRC Press. <https://doi.org/10.1201/9781003357070-10>
- Kelechi Okeke (November 29, 2017). Educating Your Customers: The Benefits & How To Achieve It. Retrieved from <https://www.cxservice360.com/educating-customers-benefits-achieve/>
- Khang, A., Chowdhury, S., & Sharma, S. (Eds.). *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.) (2022b). CRC Press. <https://doi.org/10.1201/9781003269281>
- Khang, A., Gupta, S.K., Dixit, C.K., & Somani, P., "Data-Driven Application of Human Capital Management Databases, Big Data, and Data Mining," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023a), pp. 113–133. CRC Press. <https://doi.org/10.1201/9781003357070-7>

- Khang, A., Hahanov, V., Abbas, G.L., & Hajimahmud, V.A., "Cyber-Physical-Social System and Incident Management," *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) (2022a). CRC Press. <https://doi.org/10.1201/9781003252542-2>
- Khang, A., Rani, S., Gujrati, R., Uygun, H., & Gupta, S.K. (Eds.). *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023b). CRC Press. <https://doi.org/10.1201/99781003357070>
- Khang, A., Rani, S., & Shah, V. (Eds.). *AI-Based Technologies and Applications in the Era of the Metaverse* (1st ed.) (2024). IGI Global Press. <https://doi.org/10.1201/9781668488515>
- Khanh, H.H., & Khang, A., "The Role of Artificial Intelligence in Blockchain Applications," *Reinventing Manufacturing and Business Processes Through Artificial Intelligence* (2021), pp. 20–40. CRC Press. <https://doi.org/10.1201/9781003145011-2>
- Mandloys Digital Marketing, (2013). Content Marketing Essentials 2013. <http://www.mandloys.com/contentmarketing/>
- Misra, A., Shah, V., Khang, A., & Gupta, S.K. (Eds.). *AI-Aided IoT Technologies and Applications in the Smart Business and Production* (1st ed.) (2023). CRC Press. <https://doi.org/10.1201/9781003392224>
- Morris, G., Babasaheb, J., Khang, A., Gupta, S.K., & Hajimahmud, V.A. (Eds.). *AI-Centric Modelling and Analytics: Concepts, Designs, Technologies, and Applications* (1st ed.) (2023). CRC Press. <https://doi.org/10.1201/9781003400110>
- Oppflow (2023). 4 Reasons Your Business Should Use Content Marketing. Retrived on 2023 from <https://www.linkedin.com/pulse/4-reasons-why-your-business-needs-content-marketing-opppflowio/>
- Pulizzi, J., & Barrett, N., *Get Content, Get Customers: Turn Prospects into Buyers with Content Marketing* (1st ed.) (2009). McGraw-Hill. https://escherman.files.wordpress.com/2008/02/get_content_get_customers_ebook_102407.pdf
- Rana, G., Khang, A., Sharma, R., Goel, A.K., & Dubey, A.K. (Eds.). *Reinventing Manufacturing and Business Processes Through Artificial Intelligence* (2021). CRC Press. <https://doi.org/10.1201/9781003145011>
- Rani, S., Chauhan, M., Kataria, A., & Khang, A., "IoT equipped intelligent distributed framework for smart healthcare systems," *Networking and Internet Architecture*, 2021, <https://doi.org/10.48550/arXiv.2110.04997>
- Snehal, M., Babasaheb, J., & Khang, A., "Workforce Management System: Concepts, Definitions, Principles, and Implementation," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023), pp. 1–13. CRC Press. <https://doi.org/10.1201/9781003357070-1>
- Tailor, R.K., Pareek, R., & Khang, A. (Eds.). "Robot Process Automation in Blockchain, *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.) (2022), pp. 149–164. CRC Press. <https://doi.org/10.1201/9781003269281-8>
- Vrushank, S., Khang, A., & Rani, S. (Eds.). *AI-Based Technologies and Applications in the Era of the Metaverse* (1st ed.) (2023). IGI Global Press. <https://doi.org/10.1201/9781668488515>

19 Linkages between Critical Success Factors and Factors of Workforce Performance in Remanufacturing Industry

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CONTENTS

19.1	Introduction	312
19.1.1	Origin and Overview of Remanufacturing	312
19.1.2	Industry 4.0 and Remanufacturing	312
19.1.3	6Rs Rule (Reduce, Reuse, Remanufacture, Recycle, Redesign, and Recover).....	313
19.1.4	Objective and Organization of Chapter	314
19.2	Related Work	314
19.2.1	Remanufacturing Is a Product Reprocessed in an Industrial Process.....	314
19.2.2	Remanufacturing Is an Industrial Process	315
19.2.3	Remanufacturing Is an Ultimate Form of Recycling.....	315
19.3	Critical Success Factors and Workforce Performance Measure Factors.....	316
19.3.1	Top Management Involvement and Commitment.....	318
19.3.2	Remanufacturing Process Planning.....	318
19.3.3	Forward/Reverse Logistics System.....	319
19.3.4	Remanufacturing Process Design.....	319
19.3.5	Supplier's Involvement for Remanufacturing	319
19.3.6	Sustainable Development Practices	319
19.3.6.1	Financial Performance.....	320
19.3.6.2	Productivity Performance.....	320
19.3.6.3	Remanufacturing Process Performance	320
19.3.6.4	Employees' Satisfaction.....	320
19.3.6.5	Customers' Satisfaction	321

19.4 Methodology 321

19.5 Analysis and Results 321

 19.5.1 Interpretation of Critical Factors Affecting the
 Remanufacturing Industry 321

 19.5.2 Interpretation of Workforce Performance Measure Factors
 Affecting the Remanufacturing Industry 324

19.6 Conclusion 326

References 327

19.1 INTRODUCTION

19.1.1 ORIGIN AND OVERVIEW OF REMANUFACTURING

Awareness of environmental problems related to the manufacturing industry is continuously growing worldwide. This has created tremendous pressure to reuse, remanufacture, and recycle the discarded products. In the present economic situation, organizations are being pressurized by governments to use the natural resources efficiently (Khang et al., 2023a).

In fact, many corporate units have seriously initiated product recovery management in their business plans. Today remanufacturing is being practiced in various industrial sectors and considered as very effective means of product recovery for sustainable development (Babasaheb et al., 2023a).

Remanufacturing has come up as an option which provides effective way not only to use and keep the value of the product but also extend the life of the product. It saves the energy for conversion of raw materials into finished components and 20–40% of the energy is required to manufacture a new product.

A study of automotive components has clearly demonstrated that about 85% of energy can be preserved through remanufacturing process (Fegade et al., 2015). The profit from remanufacturing activities is about 20%, although for automotive sector it goes up to 30 ± 10%.

In some cases, remanufactured products can be twice as profitable compared to the new product. The annual energy savings from remanufacturing in US alone has been estimated to be equal to electricity-generated nuclear power plants. Remanufacturing also reduces up to 0.8 million tons of CO2 emission per annum. It helped preserve the ecosystem by significantly reducing environmental pollutants caused by incineration and landfills of used products (Snehal et al., 2023).

19.1.2 INDUSTRY 4.0 AND REMANUFACTURING

Nowadays, industrial production depends on the market requirement. It consists of global competition and need have fast adopted the new technology and challenges. An Industry 4.0 concept is widely used in Europe and particularly in Germany’s manufacturing world (the Fourth Industrial Revolution, 4IR, or Industry 4.0, conceptualizes rapid change to technology, industries, and societal patterns and processes in the 21st century due to increasing interconnectivity and smart automation). Industry 4.0 involves the system of manufacturing processes,

which depends on instruments autonomously communicating with each other (Babasaheb et al., 2023b).

These technologies in Industry 4.0 along with the value chain, a model of future, i.e., smart industry which takes decentralized decisions based on self-organizing mechanisms (Khang et al., 2022d), create a virtual copy of the physical world. In Industry 4.0, a computer-driven system executes the physical processes. This is referred to as the Fourth Industrial Revolution.

The automobile industry has been affected by the lack of computer chips, which are critical automotive components produced primarily in Asia such as the time post COVID-19 pandemic. Automobile firms across the world have been forced to shut down assembly lines due to a lack of semiconductors (Gupta et al., 2023).

As a result, prices of all the goods have been hiked post-lockdown. In such situations, e-manufacturing can be a savior. Firms can use resources from the end-of-life (EOL) (in the context of manufacturing and product lifecycles is the final stages of a product's existence) stage components and store them, which can be used for manufacturing components.

Currently, there is no systematic framework for remanufacturing and it correlates to Industry 4.0 so that can be clearly identified in both practice and academic research to guide future deployments (Khang et al., 2023b). This chapter is motivated by these circumstances and provides a concept framework for remanufacturing, which enhances Industry 4.0 (Jain et al., 2023).

19.1.3 6Rs RULE (REDUCE, REUSE, REMANUFACTURE, RECYCLE, REDESIGN, AND RECOVER)

Many industries across the world are engaged in adopting remanufacturing to improve their financial and productivity performance (Sonawane et al., 2023).

The 6Rs approach is as follows.

- **Reduce** – This means to reduce the environmental impact, carbon emission, and waste, as well as minimize energy and natural resources consumption.
- **Reuse** – “Reuse” is the action or practice of using an item, whether for its original purpose or to fulfill a different function. It should be distinguished from recycling, which is the breaking down of used items to make raw materials for the manufacture of new products.
- **Remanufacture** – An example of remanufacturing is the use of a car by replacing its engine, reconditioning a refrigerator and changing its compressor, or retrieval and remanufacture of rocket motor casings in reusable space shuttles.
- **Recycling** – The extraction of a product's raw materials used in new products. An example of recycling is recycling of plastics or melting worn-out metal components.
- **Redesign** – To change the way something looks, is made, or works.
- **Recover** – To recover something is to find or get back the use of something lost or taken away.

19.1.4 OBJECTIVE AND ORGANIZATION OF CHAPTER

The available literature and various reports show that there are many sectors for which remanufacturing is being carried out presently. These include diesel engines, computers, mobile phones, refrigerators, tires, and batteries. Among these, the automotive industry is the major contributor.

The objective of research study is to frame and analyze the relationship between critical remanufacturing factors and workforce performance measure factors to support remanufacturing by development of a framework for manufacturing industry (Rani et al., 2021).

The authors extracted factors and measures through detailed literature. After prioritizing critical success factors and its sub-factors, the authors analyzed relationships through analysis and linked them with workforce performance measure factors.

The authors thus extended critical success factors on based analysis toward achieving the remanufacturing framework. The authors covered a wide spectrum of manufacturing plants in India and carried out survey-based analysis for data collection (Bhambri et al., 2022).

The rest of this chapter is organized as follows: [Section 19.2](#) describes the previous research on remanufacturing and its challenges. Various authors describe the definitions of remanufacturing. This section discusses critical success factors and sub-factors, and importance of remanufacturing. The identification of critical success factors for remanufacturing is presented in [Section 19.3](#).

Next, [Sections 19.4](#) and [19.5](#) cover the research approach used in this study and data collection. This is followed by data analysis and results. [Section 19.6](#) covers the discussion, findings and conclusion, respectively, discusses limitations of study, and indicates directions for future research (Rani et al., 2023).

19.2 RELATED WORK

Much literature is available about remanufacturing, which has been explained by different authors. Previously published articles have not defined a comprehensive set of requirements for remanufacturing or critical factors of remanufacturing industry. This section addresses the gap for the remanufacturing industry in particular.

This review discussed on the results of a survey about critical success factors and workforce performance factors. There are several definitions of remanufacturing.

19.2.1 REMANUFACTURING IS A PRODUCT REPROCESSED IN AN INDUSTRIAL PROCESS

In this process, a used product passes through a number of remanufacturing steps. These steps are inspection, cleaning, disassembly, part reprocessing, reassembly, and testing (Sundin et al., 2008). Remanufacturing has been performed in some form for decades, but it is an industrial activity. It is mainly in specific industrial fields and a few geographical areas.

There are three main tasks for a remanufacturer (Parker & Butler, 2012):

- The collection of cores.
- The remanufacturing process.
- The redistribution of remanufactured products.

Lindahl et al. (2006) have described the environmental issues within the remanufacturing industry. Remanufacturing is one form of product end-of-life strategy.

19.2.2 REMANUFACTURING IS AN INDUSTRIAL PROCESS

In this process used/broken-down products or components are restored to useful life (Ostlin, 2008). Remanufacturing is a process of restoring a non-functional, discarded, or traded-ins product. These products after processing are good as new (Lund, 2010). Remanufacturing is a “win-win-win” situation.

The customers provide fewer products for remanufactured components. Remanufacturing industries can take more profit, and the environment benefits also (Lundmark et al., 2009). The final conclusion is that remanufacturing contributes three dimensions of sustainability: environmental, financial, and societal. It saves material resources and energy resources, and it reduces waste. It creates skilled jobs and produces substantial savings for customers (Fegade et al., 2020).

Lund (2010) has described several societal benefits of remanufacturing. It makes products more widely available at lower prices. This contributes to conservation of materials and energy. It provides employment, skill acquisition, and expertise. It also provides local employment and training. Mukherjee et al. (2010) have analyzed the different issues related to remanufacturing in Indian industry.

19.2.3 REMANUFACTURING IS AN ULTIMATE FORM OF RECYCLING

In recycling, large amounts of energy and labor are wasted. But through remanufacturing, materials and energy can be saved and less waste is produced. Recent years, circular economy has been defined as a modern industrial system. Because it is easy restorative or regenerative by intention and design. As a result, it has attracted increasing interest in industry and society.

Remanufacturing is one method for a circular economy. For instance, Lind et al. (2014) have proposed remanufactures of automotive components in Europe. As a result, they can manage inter-organizational relationships in the reverse supply chain to identify which parameters are important to determine the efficiency of the relationships with suppliers.

Other instance, Mittal et al. (2014) have developed a structural model. This model is helpful for environmentally conscious manufacturing drivers. Karvonen et al. (2015) have studied remanufacturing, focusing on two areas:

- Promoting remanufacturing through collaboration networks.
- Assessing the sustainability effects of remanufacturing. This area supports the identification of future actions to widen the application of remanufacturing.

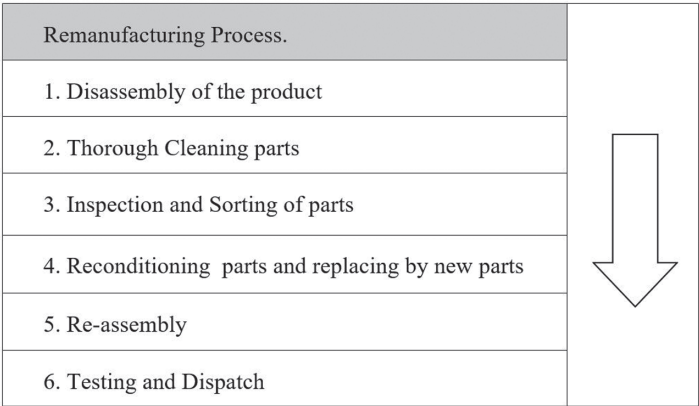


FIGURE 19.1 The path of remanufacturing process.

Ehie et al. (2016) have also analyzed the sample survey of manufacturers in developing countries as shown in [Figure 19.1](#).

19.3 CRITICAL SUCCESS FACTORS AND WORKFORCE PERFORMANCE MEASURE FACTORS

Achanga et al. (2006) have presented critical remanufacturing factors. These factors have constituted a successful implementation of lean manufacturing within manufacturing industries. [Table 19.1](#) represents identified critical remanufacturing factors. [Table 19.2](#) shows workforce performance measure factors of remanufacturing.

The gaps have identified through discussion with different level of Industrialist and academicians, from many visits to auto industries and manufacturing industries, as well as from many available in related work on 6-R rule and remanufacturing. As a result, the literature on various aspect of remanufacturing was reviewed, and a gap existing in the literature was identified. Related work shows that there is a need to focus on qualitative and quantitative study.

The different stakeholder views toward remanufacturing were considered to develop the remanufacturing industry model, which provides a simple way to implement remanufacturing practices (Shrivastava et al., 2016). The described factors are facing lot of problems in remanufacturing. There is a lack of empirical studies in the manufacturing industry.

Critical success factors (CSFs) are the vital input factors that will drive a good remanufacturing industry. The authors focus their work on critical success factors or variables of remanufacturing, as shown in [Table 19.1](#).

The description of the seven identified CSFs of good remanufacturing industry is as follow.

TABLE 19.1
Identified Critical Factors for Remanufacturing

Authors/Critical Success Factors	Top Management Involvement and Commitment	Human Resource Management	Remanufacturing Process Planning	Forward/Reverse Logistics System	Remanufacturing Process Design	Suppliers for Remanufacturing	Sustainable Development Practices
Sundin et al., 2008; Shrivastava et al., 2006; Mitsutaka et al., 2016; Khan et al., 2022	*	*	*	*	*		
Lund, 2010; Iris et al., 2015; Hartwell et al., 2016; Haleem et al., 2021		*	*	*	*		*
Achanga et al., 2006; Lind et al., 2014;	*		*		*	*	*
Seth et al., 2008; Holt et al., 2009			*	*		*	*
Ijomah et al., 1999; Ahemad & Shrivastava, 2016; Matsumoto et al., 2020	*	*				*	*

Table 19.1 depicts the critical success factors or variables of remanufacturing.

TABLE 19.2**Identified Workforce Performance Measuring Factors for Remanufacturing**

Authors/Performance Measure Factors	Financial Performance	Productivity Performance	Remanufacturing		Customers Satisfaction
			Quality Performance	Employee Satisfaction	
Shrivastava et al., 2006; Ahemad et al., 2013; Ehie et al., 2016; Mitsutaka et al., 2016; Singhal et al., 2019		*	*	*	*
Achanga et al., 2006; Iris et al., 2015; Hartwell et al., 2016	*	*	*		*
Lind et al., 2014; San-Francisco et al., 2020	*	*	*		*
Seth et al., 2008; Kalpande et al., 2021	*	*	*	*	*
Ijomah et al., 1999; Ahemad et al., 2013	*	*	*		

19.3.1 TOP MANAGEMENT INVOLVEMENT AND COMMITMENT

Top management understands the philosophy of remanufacturing. It believes that remanufacturing can be beneficial for a company in many ways. It ensures that there are adequate budgetary and resource provisions for remanufacturing, and it is ready to accept changes to facilitate remanufacturing.

Human resource management: In this factor, employees' involvement in remanufacturing is necessary. They should understand about elimination of waste. The employees' evaluation and suggestions are important for remanufacturing. With the help of evaluation, give them noticeable rewards. The recognition system is employed for employees to generate innovative ideas for remanufacturing (Hahanov et al., 2022).

19.3.2 REMANUFACTURING PROCESS PLANNING

Remanufacturing Process Planning is the factor that involves customer's willingness and convenience to return the used products, customer decides on how and when to return the used products, evaluation of return used product, facilitation for procurement of used product, and forecasting status for product returns. These used products are supplied by the customer for remanufacturing. Remanufacturing has opened new sources of profits through new market opportunities.

Inventory management covers the value of used products, which decrease during the time in inventory. Optimum inventory is utilized for the remanufacturing process. The supply of used products is greater/lesser than the demand of remanufactured products. Components are necessary between disassembly, reprocessing, and reassembly (Tailor et al., 2022).

19.3.3 FORWARD/REVERSE LOGISTICS SYSTEM

This factor covers marketing/purchasing strategies for attracting used product. The material flow is organized for components that cannot be remanufactured. The most important characteristic is lower cost for transportation or fast deliveries. The size of the possible target area depends on the cost of used product returns.

A general cost for logistics activities is needed in remanufacturing. The factor includes less lead time for delivery in the supply chain management. The used products are stored when remanufacturing. The average distance is used in transportation process.

19.3.4 REMANUFACTURING PROCESS DESIGN

Remanufacturing Process Design is the factor includes production layout and the material flow for remanufacturing, volume of used product in remanufacturing, energy demanding operations in the remanufacturing process, and remanufacturing volume is increase in the production operation.

The fixed and variable costs are linked with their processes and different remanufacturing industries. The customers' demands from the ordering lead time, supply gap in remanufacturing process.

The mobility of the equipment used in the remanufacturing process, material handling in the material flow (disassembly), material handling in the material flow (reassembly), and also material handling in the material flow (reprocessing).

The other flexibility factors involve remanufacturing process planning is also performs from order to delivery, disassembly orders set in respect to quantities for remanufacturing, reprocessing orders have set in respect to quantities for remanufacturing, and orders for new replacement component issued for remanufacturing.

Reassembly orders are set in respect to quantities for remanufacturing. Information technology software is available to manage these issues. Forecasts for the material recovery rate are used in the planning process. Incoming quality of used products should influence the capacity planning. Flexibility is necessary for operational staff in the remanufacturing process.

19.3.5 SUPPLIER'S INVOLVEMENT FOR REMANUFACTURING

This factor includes suppliers' ISO (International Organization for Standardization) series certification for remanufacturing, environmental partnership with suppliers, and OEMs (Original equipment manufacturers) for remanufacturing. The suppliers training and involvement is required for remanufacturing.

19.3.6 SUSTAINABLE DEVELOPMENT PRACTICES

Sustainable Development Practices is the factor covers environmental compliance through remanufacturing improves a firm's competitiveness, government support for remanufacturing industry is necessary and also strongly support from regional environmental regulations, creating product whose design or composition, usage minimize adverse impact on environment, reducing waste and scrap, and scientific

management of all types of wastes is critical. It is reduced the global warming and safety of environment which is beneficial for sustainable development practices.

Performance improvement factors are the vital effective factors that will drive a good remanufacturing industry. The authors highlight their work on performance measurement factors and variables of remanufacturing as shown in [Table 19.2](#).

A description of the five identified performance improvement factors of remanufacturing industry follows below.

19.3.6.1 Financial Performance

This factor covers how remanufacturing comes to be a more responsible and profitable way to do business, along with how efficient use of financial resources, technological knowledge, and operations to implement remanufacturing practices lead to competitive advantage.

Remanufacturing opens new sources of revenue through new market opportunities and cost savings – for example, expensive raw material, difficult production operations, values the components have on material and production planning, and components valued during the remanufacturing process.

19.3.6.2 Productivity Performance

Thus factor involves remanufacturing leading to higher productivity and profitability. Remanufacturing leads to cost savings and competitiveness, better resource utilization, and waste minimization. It ultimately reduces operating cost, enhances profit and productivity, promotes a better image of company, changes the organization culture, and enhances employee satisfaction, morale, productivity, and safety.

Remanufacturing improves product quality at lower cost, improves public image, and minimizes cost for materials purchasing and energy consumption.

19.3.6.3 Remanufacturing Process Performance

This factor covers technical upgrades, quality defects from new production, return of components for prolonged life of a product, lack of integrated information system and inadequate strategic planning, lack of support and guidance from regulatory authorities, lack of preparedness on the part of suppliers along with commitment resistance to change and adopt innovation, restrictive company policies toward the product/process, and transportation damage discovered in the supply chain.

A reverse logistic program reduces the consumption of virgin material. It increases the amount of goods delivered on time, scrap rate, investment, operational cost, and training cost. Between the forecasted cost of remanufacturing and the actual outcome, uncertainty lies in the timing of returns.

19.3.6.4 Employees' Satisfaction

This factor involves employment generation through remanufacturing, higher employee commitment and involvement in remanufacturing, a multiskilled and capable workforce for remanufacturing, and employee efficiency for remanufacturing.

19.3.6.5 Customers' Satisfaction

This factor is very important. It covers customer's complaints or claims about remanufactured product, and overall customer satisfaction (value for the money spent) about a remanufactured product. A more rapid method is required for delivering products to customer.

Cooperation with customers for using less energy during product transportation, when comparing the time between order and delivery of remanufactured products compare, with new products time to delivery, the purchasing and marketing will improve the organization's environmental performance and reduce overall cost, weight, and energy.

19.4 METHODOLOGY

The research methodology of establishing a relationship between critical success factors and workforce performance factors in the form of modeling assesses the impact of remanufacturing on organizational performance both in qualitative and quantitative aspects (Toke et al., 2018, 2019). Therefore, it is necessary to develop a new research instrument. Shrivastava et al. (2006) developed their instruments for data collection, based on their own research purposes, their instruments differed from each other.

After examining the instruments developed by researchers related to remanufacturing, it was observed that none of the questionnaires fully met the requirement of this research. Therefore, it was necessary to develop new research instruments.

However, the instruments developed by these researchers did give some insight into developing the instrument required for this research purpose. The Likert scale was used for collecting the data (Hajimahmud et al., 2022).

In this chapter, a quantitative method is applied for data collection along with a detailed plan for obtaining a sample from a given population quantitative method is employed for measuring the quantity or amount of a particular phenomenon by the use of statistical analysis. Mean, standard deviation, etc. can be computed for quantitative methods. Qualitative methods are a nonquantitative type of analysis which is aimed at finding the quality of a particular phenomenon (Khang et al., 2022a).

The sample is collected from organizations in Indian industries. The continuous efforts by the authors and the support of the study organizations generated a 49.2% response rate, which was quite encouraging.

The data was analyzed using software packages. The factor analysis was performed by principal component analysis using Varimax rotation. The test for significance was used to test the relationships.

19.5 ANALYSIS AND RESULTS

19.5.1 INTERPRETATION OF CRITICAL FACTORS AFFECTING THE REMANUFACTURING INDUSTRY

The factors described are some of the problems facing the remanufacturing industry. Critical success factors are the vital input factors that will drive a good remanufacturing industry. These factors are described according to the industrial point of view.

The principal component method is used. The initial number of factors is the same as the number of variables used in the factor analysis. For the analysis, 56 attributes are considered, which covers seven input factors.

Top management involvement and commitment, human resource management, remanufacturing support systems, remanufacturing process, supplier's involvement for remanufacturing, forward/reverse logistics system design, and sustainable development practices are the critical factors considered for the analysis which affects the huge impact in establishing remanufacturing industry.

As we have a larger number of attributes for the analysis, they reduced that to 12 by extracting the common variance and put in the common score. This common score can be used for the further analysis.

Initial eigenvalues and the percentage variance for these extracted variables are shown in [Figure 19.2](#). Rotation is done to maximize the variance among the attributes which were already received from the principal component analysis.

This indicates how shared variance in rotation discretely explains attribute correlation with principal components (input factors). Sum of square (SS) loading is carried out to find the correlation between the attributes and the input factors as shown in [Figure 19.2](#).

Varimax rotated factor analyses of all seven remanufacturing critical factors are shown in [Table 19.3](#), and well as the values are represented in [Figure 19.3](#).

In [Table 19.3](#), Factor I accounting for 24.5% of common variance is named as top management involvement and commitment. It assures the belief and understanding of the remanufacturing for managing economic and environmental benefits of the process. It has maximum common variance, which means involvement of the management is so crucial.

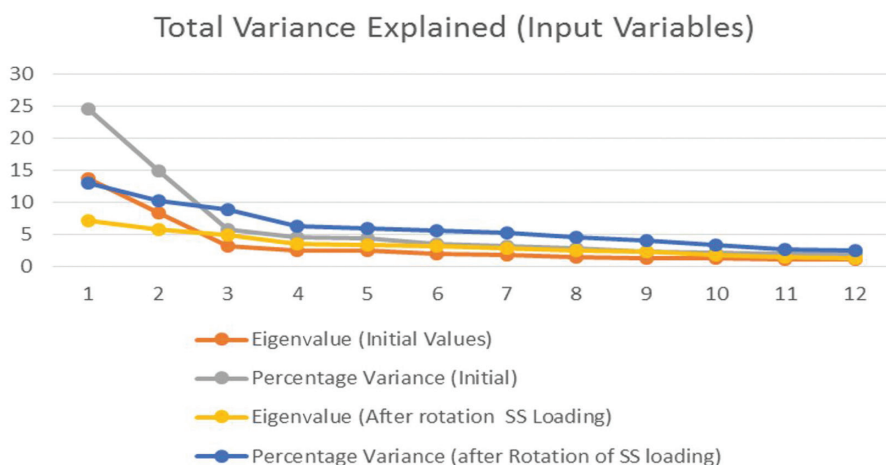


FIGURE 19.2 Total variance explained for input factors.

TABLE 19.3
Varimax-Rotated Factor Matrix on Sampled Data
(Remanufacturing Factors)

Factor	Eigenvalue	% of Variance	Cumulative %
I	13.738	24.531	24.531
II	8.336	14.886	39.418
III	3.240	5.786	45.203
IV	2.605	4.651	49.855
V	2.457	4.388	54.243
VI	1.997	3.567	57.809
VII	1.751	3.127	60.936

Willingness of management in the remanufacturing industry always plays an important role. Factor II, accounting for 14.88% of common variance, is human resource management. The workforce in the organization should also know the importance of product recovery and how to eliminate waste from the product. This factor is rated second highest because after management, these people are actively involved in the remanufacturing process. Performance of the

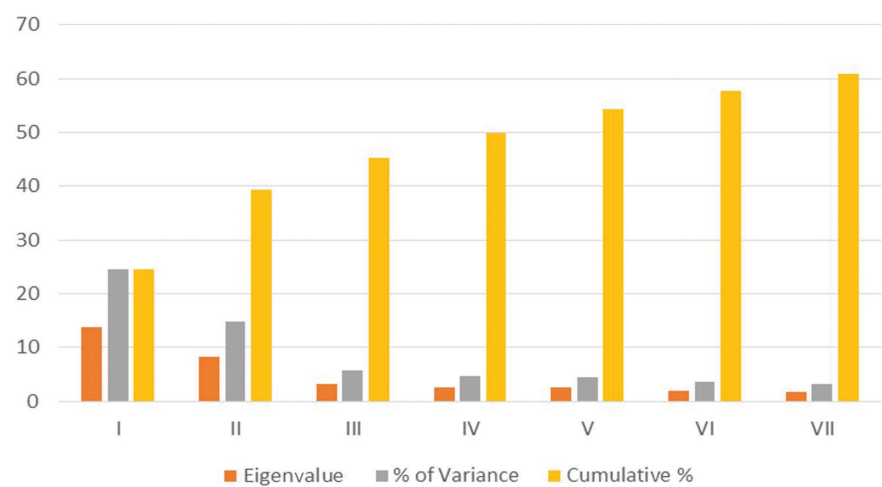


FIGURE 19.3 Varimax-rotated factor matrix values for critical success factors.

remanufactured product and efficiency of the process are completely dependent upon these categories.

Factor III, accounting for 5.78% of common variances, is the remanufacturing support system. This factor describes the customer's willingness to purchase remanufactured products, as well as how and when to return the used products, forecasting done for product returns, facilitation for procurement of used products, and evaluation of returned products.

Factor IV, accounting for 4.65% of common variance, is the remanufacturing process. This proves that the efficiency of the remanufacturing process is also one of the important factors to gain the market. The manufacturer can overcome the phycology of the customer by providing good finished and workable remanufacturing product against the newly manufactured product.

Factor V accounts for 4.39% of common variance; this factor is forward/reverse logistics. It focuses upon the logistic issues in the process. Delay in logistics may affect the cost of the remanufactured product.

Factor VI accounts for 3.57% of common variance; this factor is suppliers' involvement in remanufacturing. It involves suppliers' ISO series certification for remanufacturing. Factor VII, accounting for 3.12% of common variance, is sustainable development practices.

19.5.2 INTERPRETATION OF WORKFORCE PERFORMANCE MEASURE FACTORS AFFECTING THE REMANUFACTURING INDUSTRY

For the interpretation of the workforce performance measures, financial performance, productivity performance, remanufacturing quality performance, customer satisfaction, and employee satisfaction are considered. Attributes are set considering these performance indicators for the collection of the data.

A total of 49 questions based on these factors were used to collect the data. Similarly, a principal component method was used for the extraction of the important attributes focusing on the performance measures.

Initial eigenvalues and the percentage variance of these extracted attributes are derived; then the sum of the square loading is carried out for the correlation of the attributes. Change in the eigenvalues and percentage variance after rotation are shown in [Figure 19.4](#).

The Varimax-rotated matrix shown in [Table 19.4](#) clearly indicates that financial performance has maximum weights in the workforce performance of the remanufacturing process. It shows that 40.35% common variance means almost two thirds of the cumulative percentage. It also clear that if any process is not financial viable, no industry will be willing to stay in the business.

A remanufacturing process is an economical process but the workforce performance of the process depends upon many dynamic factors related to market and technology.

Productivity performance is the second most important performance measure. It covers 6.55% of the common variance. The success of the process always depends upon the productivity of the organization (Khang et al., 2022c).

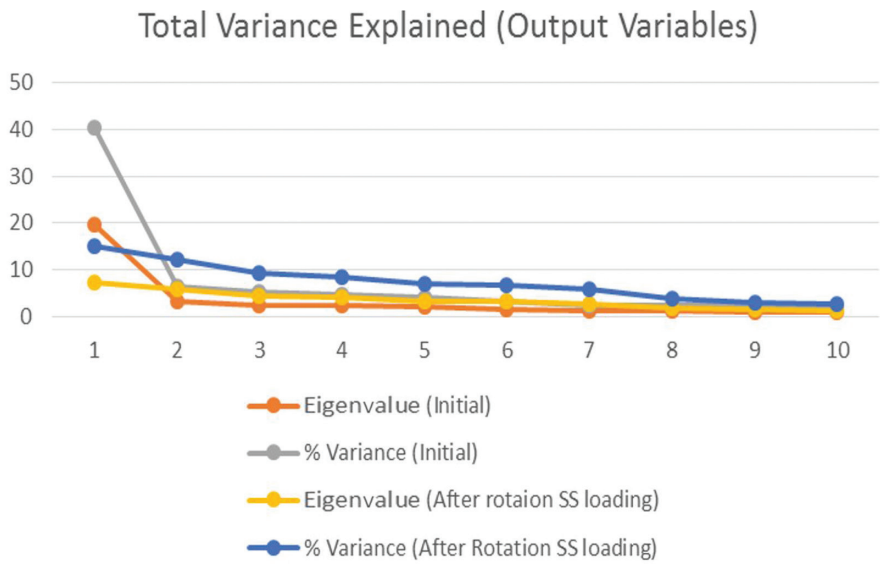


FIGURE 19.4 Total variance for workforce performance measures.

Remanufacturing quality is also one of the important measures which indicates at what quality the product is recycled and further decides the reliability of the remanufactured product. Accountability of this measure has 5.308% of common variance.

The fourth and fifth workforce performance measures are employee satisfaction and customer satisfaction. Both have the common variance of nearly 4 percent. Satisfaction level of the customer and employee can be enhanced by counseling programs which focusing the importance of the process and its environmental impact on society, as [Figure 19.5](#) shows.

TABLE 19.4
Varimax-Rotated Factor Matrix on Sampled Data
(Remanufacturing Workforce Performance Measures)

Factor	Eigenvalue	% of Variance	Cumulative %
I	19.774	40.356	40.356
II	3.213	6.557	46.913
III	2.601	5.308	52.220
IV	2.394	4.886	57.106
V	2.119	4.324	61.430

Table 19.4 depicts the maximum weights in the workforce performance of the remanufacturing process.

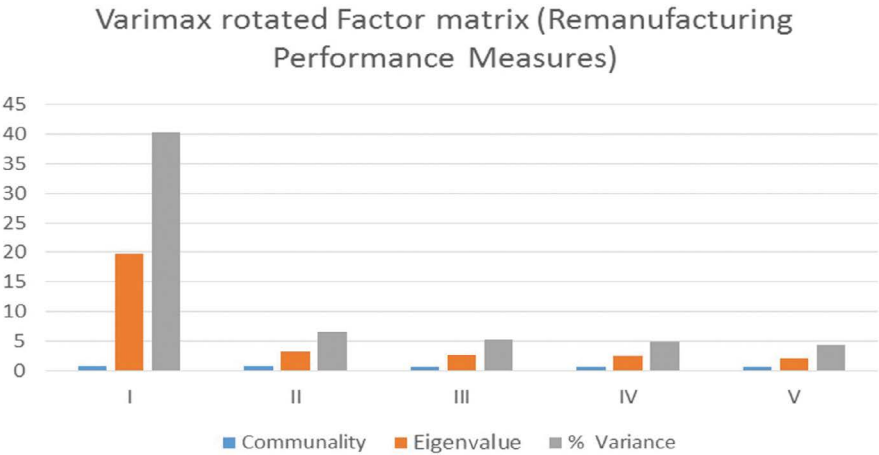


FIGURE 19.5 Varimax-rotated factor matrix values for workforce performance measures.

19.6 CONCLUSION

Many empirical studies have been carried out in remanufacturing. This chapter focused on linkage of critical success factors and workforce performance measurement factors for the remanufacturing industry. In general, a model is needed to develop the relationship between new products and returned products. The chapter discussed this gap in particular for the remanufacturing industry. Responses were collected from various manufacturing industries in India (Rana et al., 2021).

The statistical tests were applied to the data with the help of software, and the results were discussed. With the help of results, we have prepared the structural coefficient model. We found that there is no relationship between remanufacturing critical factors and economic performance or productivity.

This study has the potential to contribute to and improve financial and productivity performance. Industry 4.0 involves a wide set of technologies that provide a good roadmap for innovations and creative solutions (Khanh et al., 2021).

This work is limited to manufacturing industries in India. Collection of samples from stakeholders can give mixed perceptions, which will affect results. However, the respondents were educated to remove the bias and get accurate results (Misra et al., 2023).

In India in particular and other countries in general, there are very few reported studies, especially for remanufacturing implementation. However, all efforts were made to remove the bias in perception of the respondents by thoroughly explaining to them the context of the questionnaire and educating them, Still, for remote respondents, the possibility of biased response (although very small) cannot be ruled out (Morris et al., 2023).

India needs to take a few steps to ensure remanufacturing success in Industry 4.0. That means skill development to develop a robust data security environment (Khang et al., 2022b). The challenges in Industry 4.0 involve training, type of process and

work organization, lack of specialist staff, effective plant layout, and strong network infrastructure (Vrushank et al., 2023).

REFERENCES

- Achanga, P., Shahab, E., Roy, R., & Nelder, G. (2006). Critical success factors for lean implementation within SMEs. *Journal of Manufacturing Technology Management*, 17(4), 460–471. <https://www.emerald.com/insight/content/doi/10.1108/17410380610662889/full/html>
- Ahemad, M., & Shrivastava, R.L. (2013). A Study of Finding Impact of Green Manufacturing Practices on Firms Profitability. *Proceeding of International Conference on Industrial Engineering in S.V.N.I.T. Surat*, pp. 308–313. <https://www.sciencedirect.com/science/article/pii/S0959652616310034>
- Babasaheb, J., Sphurti, B., & Khang, A. (2023a). “Industry Revolution 4.0: Workforce Competency Models and Designs,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches*, (1st ed.), pp. 14–31. CRC Press. <https://doi.org/10.1201/9781003357070-2>
- Babasaheb, J., Sphurti, B., & Khang, A. (2023b). “Design of Competency Models in the Human Capital Management System,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 32–50. CRC Press. <https://doi.org/10.1201/9781003357070-3>
- Bhambri, P., Rani, S., Gupta, G., & Khang, A. (2022). *Cloud and Fog Computing Platforms for Internet of Things*, CRC Press. <https://doi.org/10.1201/9781032101507>
- Ehie, I., & Muogboh, O. (2016). Analysis of manufacturing strategy in developing countries: A sample survey of Nigerian manufacturers. *Journal of Manufacturing Technology Management*, 27(2). <https://www.emerald.com/insight/content/doi/10.1108/JMTM-07-2014-0094/full/html>
- Fegade, V., Shrivastava, K., Kale, A.V., & Shrivastava, R.L. (2020). Remanufacturing feasibility of bike suspension by hybrid TOPSIS-Taguchi optimization. *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences*, 71(1), 125–133. <https://www.akademibaru.com/submit/index.php/arfmts/article/view/2976>
- Fegade, V., Shrivatsava, R.L., & Kale, A.V. (2015). Design for remanufacturing: Methods and their approaches. *Materials Today: Proceedings*, 2(4–5), 1849–1858. <https://www.sciencedirect.com/science/article/pii/S2214785315003752>
- Gupta, S.K., Somani, P., Dixit, C.K., Pathak, A., & Khang, A. (2023). “Data Mining Processes and Decision-Making Models in Personnel Management System,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 89–112. CRC Press. <https://doi.org/10.1201/9781003357070-6>
- Hajimahmud, V.A., Khang, A., Hahanov, V., Litvinova, E., Chumachenko, S., & Alyar, A.V. (2022). “Autonomous Robots for Smart City: Closer to Augmented Humanity,” *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003252542-7>
- Haleem, A., Khan, S., Luthra, S., Varshney, H., Alam, M., & Khan, M. (2021). Supplier evaluation in the context of circular economy: A forward step for resilient business and environment concern. *Business Strategy and the Environment*, 30(4), 2119–2146. <https://onlinelibrary.wiley.com/doi/abs/10.1002/bse.2736>
- Hartwell, I., & Marco, J. (2016). Management of intellectual property uncertainty in a remanufacturing strategy for automotive energy storage systems. *Journal of Remanufacturing*, 6, 3. <https://link.springer.com/article/10.1186/s13243-016-0025-z>
- Holt, D., & Ghobadian, A. (2009). An empirical study of green supply chain management practices amongst UK manufacturers. *Journal of Manufacturing Technology*

- Management*, 20(7), 933–956. <https://www.emerald.com/insight/content/doi/10.1108/17410380910984212/full/html>
- Ijomah, W.L., Bennett, J.P., & Pearce, J. (1999). Remanufacturing: Evidence of environmentally conscious business practice in the UK. *IEEE Transactions*, 16, 171–183. <https://ieeexplore.ieee.org/abstract/document/747607/>
- Jain, P., Tripathi, V., Malladi, R., & Khang, A., (2023). “Data-Driven AI Models in the Workforce Development Planning,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 179–198. CRC Press. <https://doi.org/10.1201/9781003357070-10>
- Kalpande, S.D., & Toke, L.K. (2021). Assessment of green supply chain management practices, performance, pressure and barriers amongst Indian manufacturer to achieve sustainable development. *International Journal of Productivity and Performance Management*, 70(8), 2237–2257. <https://www.emerald.com/insight/content/doi/10.1108/IJPPM-02-2020-0045/full/html>
- Karvonen, I., Jansson, K., Tonteri, H., Vatanen, S., & Uoti, M. (2015). Enhancing remanufacturing – Studying networks and sustainability to support Finnish industry. *Journal of Remanufacturing*, 5, 5. <https://link.springer.com/article/10.1186/s13243-015-0015-6>
- Karvonen, I., Jansson, K., Vatanen, S., Tonteri, H., & Uoti, M. (2015). Remanufacturing efficient model of circular economy, *VIT Technology*, 207, VIT Expo. <https://link.springer.com/article/10.1186/s13243-015-0015-6>
- Khang, A., Chowdhury, S., & Sharma, S. (Eds.). (2022a). *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003269281>
- Khang, A., Gupta, S.K., Dixit, C.K., & Somani, P. (2023a). “Data-Driven Application of Human Capital Management Databases, Big Data, and Data Mining,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 113–133. CRC Press. <https://doi.org/10.1201/9781003357070-7>
- Khang, A., Hahanov, V., Abbas, G.L., & Hajimahmud, V.A. (2022b). “Cyber-Physical-Social System and Incident Management,” *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003252542-2>
- Khang, A., Hahanov, V., Litvinova, E., Chumachenko, S., Hajimahmud, V.A., & Alyar, A.V. (2022c). “The Key Assistant of Smart City – Sensors and Tools,” *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.). 17 (10). CRC Press. <https://doi.org/10.1201/9781003252542-17>
- Khang, A., Rani, S., Gujrati, R., Uygun, H., & Gupta, S.K. (Eds.) (2023b). *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.). CRC Press. <https://doi.org/10.1201/99781003357070>
- Khang, A., Rani, S., & Sivaraman, A.K. (2022d). *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) CRC Press. <https://doi.org/10.1201/9781003252542>
- Khan, S., Haleem, A., & Fatma, N. (2022). Effective adoption of remanufacturing practices: A step towards circular economy. *Journal of Remanufacturing*, 12, 167–185. <https://link.springer.com/article/10.1007/s13243-021-00109-y>
- Khanh, H.H., & Khang, A. (2021). “The Role of Artificial Intelligence in Blockchain Applications,” *Reinventing Manufacturing and Business Processes Through Artificial Intelligence*, pp. 20–40. CRC Press. <https://doi.org/10.1201/9781003145011-2>
- Lind, S., Olsson, D., & Sundin, E. (2014). Exploring inter-organizational relationships in automotive component remanufacturing. *Journal of Remanufacturing*, 4(5). <https://link.springer.com/article/10.1186/2210-4690-4-5>
- Lindahl, M., Sundlin, E., & Ostlin, J. (2006). Environmental issues within the remanufacturing industry, Proceeding of 13th CIRP IC on Life Cycle Engineering, Leuven, pp. 447–452. <https://www.diva-portal.org/smash/record.jsf?pid=diva2:256350>

- Lund, R.T. (2010). Re-manufacturing: The Experience of the United States and implications for developing countries. World Bank, Washington DC. Vol. 87(2), 19–29. <https://digital-library.theiet.org/content/conferences/10.1049/cp.2010.0404>
- Lundmark, P., Sundin, E., & Bjorkman, M. (2009). Industrial challenges within the remanufacturing system, Proceeding of Swedish Production Symposium, Stockholm, Sweden, pp.132–138. <https://www.diva-portal.org/smash/record.jsf?pid=diva2:375722>
- Matsumoto, M., Chinen, K., Jamaludin, K., & Yusoff, B. (2020). “Barriers for Remanufacturing Business in Southeast Asia: The Role of Governments in Circular Economy,” *EcoDesign and Sustainability I*, Sustainable Production, Life Cycle Engineering and Management, pp. 151–161. Springer. https://doi.org/10.1007/978-981-15-6779-7_11
- Misra, A., Shah, V., Khang, A., & Gupta, S.K. (Eds.). (2023). *AI-Aided IoT Technologies and Applications in the Smart Business and Production* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003392224>
- Mitsutaka, M., Yang, S., Martinsen, K., & Yasutaka, K. (2016). Trends and research challenges in remanufacturing. *International Journal of Precision Engineering and Manufacturing Green Technology*, 3(1), 129–142. <https://link.springer.com/article/10.1007/s40684-016-0016-4>
- Mittal, K.V., & Singh, S.K. (2014). Development of a structural model of environmentally conscious manufacturing drivers. *Journal of Manufacturing Technology Management*, 25(8), 1195–1208. <https://www.emerald.com/insight/content/doi/10.1108/JMTM-02-2013-0012/full/html>
- Morris, G., Babasaheb, J., Khang, A., Gupta, S.K., & Hajimahmud, V.A. (Eds.). (2023). *AI-Centric Modelling and Analytics: Concepts, Designs, Technologies, and Applications* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003400110>
- Mukherjee, K., & Mondal, S. (2010). Analysis of issues relating to remanufacturing technology – A case of an Indian company. *Technology Analysis & Strategic Management*, 21(5), 639–652. <https://www.tandfonline.com/doi/abs/10.1080/09537320902969174>
- Ostlin, J. (2008). *Analysis and managing material flows and remanufacturing processes*, Thesis No.1192, Department of Mechanical Engineering, Linköping University. <https://ieeexplore.ieee.org/abstract/document/4222886/>
- Parker, D., & Butler, P. (2012). An introduction to remanufacturing, CRR, UK. <https://www.icevirtuallibrary.com/doi/abs/10.1680/warm.2010.163.4.141>
- Rana, G., Khang, A., Sharma, R., Goel, A.K., & Dubey, A.K. (Eds.). (2021). *Reinventing Manufacturing and Business Processes Through Artificial Intelligence*, CRC Press. <https://doi.org/10.1201/9781003145011>
- Rani, S., Bhambri, P., Kataria, A., Khang, A., & Sivaraman, A.K. (Eds.). (2023). *Big Data, Cloud Computing and IoT: Tools and Applications* (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/9781003298335>
- Rani, S., Chauhan, M., Kataria, A., & Khang, A. (2021). IoT equipped intelligent distributed framework for smart healthcare systems. *Networking and Internet Architecture* 2, 30. <https://doi.org/10.48550/arXiv.2110.04997>
- San-Francisco, A., Sopelana, N., Fernandez, J., Otegi, J., & Minguez, R. (2020). Analysis about the incorporation of remanufacturing concept into life cycle assessment theories. *Journal of Remanufacturing* 10(2), 127–139. <https://link.springer.com/article/10.1007/s13243-019-00078-3>
- Seth, D., Seth, N., & Goel, D. (2008). Application for value stream mapping (VSM) for minimization of wastes in the processing side of supply chain of cotton seed oil industry in Indian context. *Journal of Manufacturing Technology Management*, 29(4), 529–550. <https://www.emerald.com/insight/content/doi/10.1108/17410380810869950/full/html>
- Shrivastava, R.L., & Lokhande, A.D. (2016). Linkage between effort factors and organizational performance: Remanufacturing industry. *International Journal of Management and Research*, 3, 301–310. <https://www.emerald.com/insight/content/doi/10.1108/14635770510609015/full/html>

- Shrivastava, R.L., Mohanty, R.P., & Lakhe, R.R. (2006). Linkages between total quality management and organizational performance: An empirical study for Indian industry. *Journal of Production Planning & Control*, 17(1), 13–30. <https://www.tandfonline.com/doi/abs/10.1080/09537280500324265>
- Singhal, D., Tripathy, S., & Jena, S. (2019). Acceptance of remanufactured products in the circular economy: An empirical study in India. *Management Decision*, 57(4), 953–970. <https://www.emerald.com/insight/content/doi/10.1108/MD-06-2018-0686/full/html>
- Snehal, M., Babasaheb, J., & Khang, A. (2023). “Workforce Management System: Concepts, Definitions, Principles, and Implementation,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 1–13. CRC Press. <https://doi.org/10.1201/9781003357070-1>
- Sonawane, A., & Khang, A. (2023). “Challenges Faced by Marketers in Developing and Managing Contents in Workforce Development System,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.), pp. 332–359. CRC Press. <https://doi.org/10.1201/9781003357070-18>
- Sundin, E., Ostlin, J., Ronnback, A.O., Lindahl, M., & Sandstrom, G.O. (2008). “Remanufacturing of Products used in Product service System Offerings,” *Manufacturing Systems and Technologies for the New Frontier*, pp. 537–542. Springer, https://link.springer.com/chapter/10.1007/978-1-84800-267-8_110
- Tailor, R.K., Pareek, R., & Khang, A. (Eds.) (2022). “Robot Process Automation in Blockchain,” *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.), pp. 149–164. CRC Press. <https://doi.org/10.1201/9781003269281-8>
- Toke, L.K., & Kalpande, S.D. (2018). A framework of enabler’s relationship for implementation of green manufacturing in Indian context. *International Journal of Sustainable Development & World Ecology*, 25(4), 303–311. <https://www.tandfonline.com/doi/abs/10.1080/13504509.2017.1393635>
- Toke, L.K., & Kalpande, S.D. (2019). Critical success factors of green manufacturing for achieving sustainability in Indian context. *International Journal of Sustainable Engineering*, 12(6), 415–422. <https://www.tandfonline.com/doi/abs/10.1080/19397038.2019.1660731>
- Vrushank, S., Khang, A., & Rani, S. (Eds.). (2023). *AI-Based Technologies and Applications in the Era of the Metaverse* (1st ed.). IGI Global Press. <https://doi.org/10.1201/9781668488515>

20 A Study on the Impact of the Industry 4.0 on the Employees Performance in Banking Sector

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CONTENTS

20.1 Introduction	331
20.2 Related Work	333
20.3 Research Methodology	334
20.3.1 Sampling and Data Collection	334
20.3.2 Research Instrument	336
20.4 Results and Discussion	337
20.4.1 Results	337
20.4.2 Correlation and Regression Analysis	338
20.5 Discussions	341
20.6 Conclusion	341
References	342

20.1 INTRODUCTION

The banking industry has countless opportunities for innovation. It can help improve customer service, streamline branch operations, cut costs and wait times for customers, and boost employee productivity (Khang et al., 2024). We are on the verge of the Fourth Industrial Revolution, also known as Industry 4.0, right now. Industry 4.0 has accelerated due to recent advancements in information and communication technology (ICT).

Industry 4.0's innovations and technological advancements will offer a wide range of workable solutions to the expanding needs of all industries. The world's top economists believe that the Fourth Industrial Revolution is currently underway. In the past three centuries, civilization has seen three industrial revolutions. Each of these industrial revolutions was marked by technological advancements that significantly influenced the progress of all humankind (Babasaheb et al., 2023a).

A new era, which might be called the age of digital transformation, began after the process of globalization and the connection of the world into one global market, which led to an unhindered expansion of business (Snehal et al., 2023).

The new digital era's defining trait is that it constantly assumes new dimensions and forms. Banks have significantly modified their operations to reflect changes in the business climate and have as a result embraced and implemented various digitalization-related activities.

A definite indication that the banking industry has taken impending changes seriously is the ongoing process of developing new banking products and services that are directly related to the digitalization process (Tailor et al., 2022). These changes undoubtedly produce a competitive advantage and improve a company's position in the market (Babasaheb et al., 2023b).

The literature review for this chapter focused on Industry 4.0, employee performance, and the Indian banking industry. This literature review served the purposes of formulating the problem statement, outlining the research questions, identifying the factors pertinent to the issue under investigation, putting forth research hypotheses, and creating a thorough research model for the study that would be conducted (Gupta et al., 2023).

A wide range of sources, including major public search engines, handbooks, annual reviews, government documents, abstracting and indexing services, and review articles, were looked at for this purpose.

Second, a quantitative technique was employed to carry out the data analysis procedure after the literature research, which included using a matrix-based survey questionnaire to compile the participants' responses and validate the suggested hypotheses. This analysis was conducted using IBM SPSS Statistics (Khang et al., 2023a).

Senior bank personnel from Indian private banks were among the study's participants. Industry 4.0 was used as the independent variable, and five factors—quick and efficient response, safe and simple working, security and trust, timely completion of work, and job quality—were taken into account to determine how Industry 4.0 technologies affected the performance of bank personnel. The dependent variable was chosen to be the employees' performance in the banking industry.

The results revealed a strong correlation between the five criteria and the influence of Industry 4.0 technology on bank performance, i.e., the adoption of Industry 4.0 and its associated cutting-edge technologies in the banking sector seems to boost employees' performance in the banking sector.

The results of this study could be used by business service providers, Industry 4.0 specialists, academics, and researchers in many social science domains to develop a thorough understanding of the relationship between Industry 4.0 technology and the productivity of banking sector personnel (Jain et al., 2023).

The study that was conducted also concentrated primarily on the managerial perspective of Industry 4.0, which could help managers and policymakers from the banking industry make better and more informed decisions that would, over time, not only improve the overall performance of employees but also help achieve a sustainable competitive advantage in the context of Industry 4.0 (Sonawane et al., 2023).

The objective of this research work is to assess and determine the impact of Industry 4.0 and the five factors of the technology in the era of Industry 4.0 – quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality – on the employees' performance of the banking sector in India.

20.2 RELATED WORK

Starting with the first Industrial Revolution (Industry 1.0) as steam power, water power, and mechanization, technology evolves into mass production, assembly line, and electricity. Then it evolved further during the Third Industrial Revolution (Industry 3.0) and the introduction of computerization and automation. The Fourth Industrial Revolution is the move towards digitization.

Industry 4.0 is the concept of automation and data exchange in manufacturing technologies, which enable the use of the Internet of Things (IoT), cyber-physical systems (CPS) (Khang et al., 2022b), big data analytics, cloud computing, and cognitive computing to move toward a smart world (Hercko et al., 2015).

The phenomena of Industry 4.0 was first introduced in 2011 in Germany as a proposal for the development of a new concept of German economic policy based on high-tech strategies. It is based on the concepts and technologies that include cyber-physical systems, the Internet of Things (IoT), the Internet of Services (IoS), the Industrial Internet of Things (IIoT), big data analytics, etc (Rani et al., 2021).

These concepts enhance perpetual communication via the internet, allowing a continuous interaction and exchange of information not only between humans (C2C) and humans and machines (C2M) but also between the machines themselves (M2M) (Rana et al., 2021).

Nine tools can be identified as Industry 4.0 influencers to the manufacturing sector: advanced robots, additive manufacturing, augmented reality, simulation, horizontal and vertical system integration, IoT, cloud computing, cybersecurity, and big data analytics (Rani et al., 2023).

To capture benefits of the digital technologies, companies need to manage their information architecture effectively as it is critical for their success. In such integration of legacy systems, migrating to the cloud and building cyber-resilience is significant.

For the last few decades, the emergence of technology played a major role in increasing the efficiency of an organization. Since the beginning of industrialization, technological drive has created massive changes in the way of doing business; hence in retrospect, these deviations can be defined as industrial revolutions (Wilkesmann et al., 2017).

During the First Industrial Revolution, production facilities were developed with the help of water and steam power. The Second Industrial Revolution brought mass production in the manufacturing sector with the help of electrical energy. The Third Industrial Revolution introduced electronic and information technologies that furthered production automation (Hajimahmud et al., 2022).

Over the last few years, Industry 4.0 has emerged as a promising technology framework used for integrating and extending manufacturing processes at both intra-organizational and inter-organizational levels (Xu et al., 2018). The idea of the Fourth Industrial Revolution may sound good at first, but there is a multitude of challenges, risks, and barriers associated with its implementation.

Defining appropriate infrastructures and standards, ensuring data security, and educating employees are among the issues that need to be addressed on the road to Industry 4.0 (Hofmann & Rüscher, 2017).

The availability of real-time information establishes a baseline for value creation by networking all related entities onto a single platform with optimal value flow at

any time. The association between people, objects, real time and self-organizing systems are affected by the facts of cost, availability, and resource consumption. Hence the use of cloud computing and cyber-physical systems is important in the banking industry to offer better customer service (Wilkesmann et al., 2017).

Industry 4.0 innovative technologies like biometric technology help to improve the user experience of banking customers by providing a simple process to verify existing customers or to enhance “know your customer” (KYC) onboarding methods. This will eliminate the use of passbooks and passwords for login or identification purposes in the banking system.

Customers can create their own set of biometric credentials and then use a combination of these biometrics to log in, verify their accounts, and authenticate transactions (Cook, 2017).

Industry 4.0 can be identified as an emerging platform of technologies that revolutionize the rate of productivity per employee while reducing the cost of controlling and compliance incurred by corporations.

Industry 4.0 provides the banking sector with various technological opportunities, such as self-service bank terminals, chatbots, mobile banking, blockchain, and AI-based fraud detection technology (Khang et al., 2022a).

Hypotheses Development:

- H1. **Quick and efficient response** has a significant relationship with employees’ performance.
- H2. **Safe and simple working** has a significant relationship with employees’ performance.
- H3. **Security and trust** have a significant relationship with employees’ performance.
- H4. **Timely completion of work** has a significant relationship with employees’ performance.
- H5. **Work quality** has a significant relationship with employees’ performance.

Research Model: Based on the formal systematic literature review and findings and interpretations of the literature, a comprehensive research model was proposed in this study, which is shown in [Figure 20.1](#).

20.3 RESEARCH METHODOLOGY

20.3.1 SAMPLING AND DATA COLLECTION

The banking sector in India was chosen as the target sector. The target population was composed of senior-level employees from the private banks, including managers ranging from first-level positions to top-level positions, as well as subordinate employees working under the supervision of first-level managers who possess the technological know-how of Industry 4.0. The data were collected through a survey questionnaire (Khanh et al., 2021). The study has opted to use the convenience sampling.

Accordingly, 500 survey questionnaires were sent out online to the respondents, with adequate instructions on how to fill out the survey questionnaires properly, as

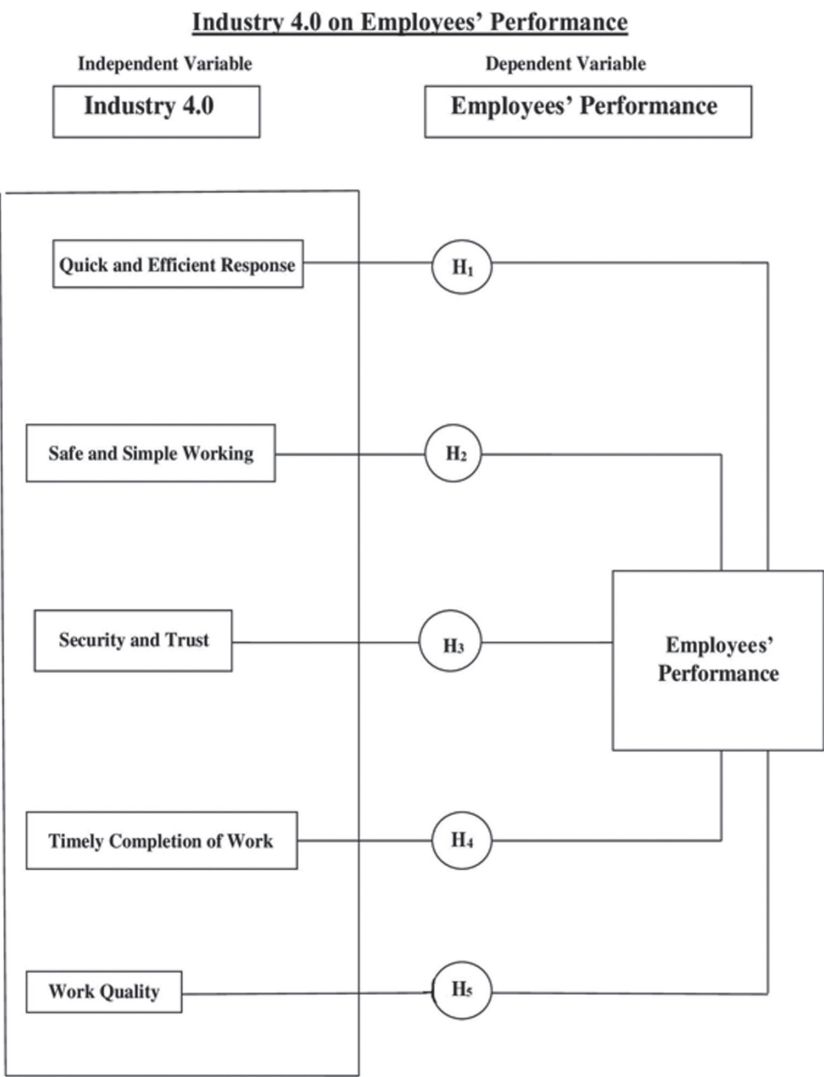


FIGURE 20.1 Research model.

Source: Author's owner.

well as the purpose of the research, which proved to be efficient. The online method of questionnaire distribution was conducted through Google forms.

After sending out the survey questionnaires online, 469 responses were collected, with a 93.8% response rate. 469 survey questionnaires were thus used finally, which is an acceptable sample size for conclusions.